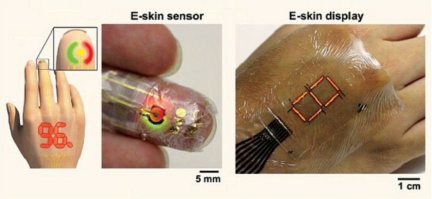

System outline of a blood oxygen level monitor: red and green PLEDs are directed to shine into the finger; reflected light from inside the finger is caught by an ultraflexible organic photodetector; this reflected light provides a measure of blood oxygen and pulse rate; the output of the sensor can be shown on a PLED display (Image credit: Tomoyuki Yokota et al./Someya Laboratory, University of Tokyo)

ever, the reliability of sensory results is highly dependent on panelists' acuity and correct application of the sensory practice. Therefore sensory attributes measurement using instrumental methods is a desirable aim. A new concept has been introduced in the sensor development using several sensors of low selectivity simultaneously.

Scientists have devised a detector that can distinguish smells better than a service dog. This sensor can detect explosive vapours and identify them. Called electronic nose, the device is able to sense and recognise traces of almost all types of explosives, from saltpetre to hexogen (which cannot be detected by even the most advanced technology).

For individual use

In the very short-term future, sight, taste and smell sensors will be available to new industries for incorporation into their own products, or for individuals directly. So, refrigerators of the future may have sensors that detect when food is going bad. These might be connected to a food-management system that alerts the user, who can then check in order to make a grocery list.

Cars may have sensors that detect carbon dioxide. The level of carbon dioxide rises in a car when driving with the windows shut. That's dangerous because carbon dioxide can make the driver sleepy. A sensor would alert the occupants to stop for fresh air.

Doctors in the future may test patients' breath to detect the onset of diseases like diabetes or cancer. A new study shows that organic compounds in exhaled breath can indicate whether a person has lung cancer—as well as its stage—and distinguish cancer from chronic obstructive pulmonary disease.

Skin is probably one of the best wearable technologies that nature ever developed. Skin on our fingertips contains pressure sensors that produce voltage pulses when we touch things. The frequency at which these pulses are created gives the brain information about what we're touching.

Prosthetic limbs could soon be covered with an electronic skin that mimics human touch. Scientists have developed a flexible plastic skin that can communicate a sense of pressure to brain cells. The plastic skin generates voltage pulses when pressure is applied. The higher the pressure, the closer the carbon nanotubes are pushed together. This increases the conductivity frequency of voltage pulses.

Electronic skin is capable of generating the same frequency of voltage pulses as human skin—up to 200 hertz. Being digital means the system is low powered, which is also important since, to mimic human touch, prosthetic skin will need to have thousands of sensors in the space of a fingertip to feel properly.

Currently electronic skin can only detect static pressure rather than moving pressure (for example, a brushing action). It is hoped that further development of a range of different sensors within five years will give artificial skin the full range of features involved in touch, including vibration, texture and temperature.

For visually impaired people, devices have been developed that can transmit the spectrum of colours and lighting around, along with spatial orientation into a mouth cavity, using slight electric stimulation.

A special electric collar can charge dogs with electric shock when these start barking too much or run too far away. So, it brings up the dogs with a sense of attachment. The dog collar may start with a vibration to warn before punishment.

Sensory punishing devices have already been developed for people. These punish the wearer with a slight electric shock in case the wearer passes a deadline, smokes after having pledged to quit or breaks some other rules established by themselves. The device is designed to facilitate the fight against bad habits.

Scientists are developing a feel-space belt that allows the wearer to feel the Earth's magnetic field and be oriented in the four winds, just like birds and bats. The belt just transforms the magnetic currents into vibrations that the body can easily perceive. In reality, the new sensation is just a cognitive effect induced by a physical impact on receptors of the old sense, which is tactility.

It is safe to say that this transition of meaning of one 'sense' via the other sense is symbolical. It requires an intermediary. Therefore it requires time and effort to recognise and learn the content of signals, while the real, natural senses are immediate for perception as these require no symbolic interpretation. We feel cold or water directly by sensors designed to feel such things.

Technological advancements

As mentioned earlier, e-nose is an artificial olfaction device with an array of chemical gas sensors, a sampling system and a pattern-classification algorithm to recognise, identify and compare gases, vapours or odours. Thus e-noses mimic the human olfactory system.

Apart from food quality detection, these devices have been successfully used in a wide variety of applications including wastewater management, measurement and detection of air and water pollution, health care and warfare. One of their strengths is that the data gathered can be interpreted without bias.